MODEL LAB · 2026-10-12

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

28,490
held-out plate appearances

3.05%
held-out HR rate

0.5961
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-01, scored on the 28,490 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5779	1.61×	0.02952	0.13559	0.00491
KNN comps Prior is the pooled rate of the nearest comparables.	0.5961	1.45×	0.02948	0.13468	0.00474
SVR Support-vector regression on the same feature matrix.	0.5887	1.43×	0.02953	0.13539	0.00646
Random forest Forest regression; a learned similarity metric.	0.5898	1.45×	0.02951	0.13508	0.00618
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5961	1.45×	0.02948	0.13468	0.00474
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5960	1.47×	0.02948	0.13468	0.00479
Linear regression Ordinary least squares on the same features.	0.5904	1.42×	0.02951	0.13512	0.00532
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5910	1.50×	0.02952	0.13522	0.00651
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5899	1.46×	0.02951	0.13512	0.00662
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5907	1.42×	0.02952	0.13520	0.00644

BLEND WEIGHTS (FOREST VARIANT)

KNN **0.90** SVR **0.00** RF **0.10**

SHRINKAGE K PER PRIOR

league **249** knn **249** svr **249** rf **249** core **249** ensemble **249** linear **249** logistic **249** nn **249** xgb **249**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,116 PA, actual HR rate 2.37%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5823	1.48×	0.02319	0.11242	0.01187
KNN comps Prior is the pooled rate of the nearest comparables.	0.5746	1.08×	0.02323	0.11256	0.01296
SVR Support-vector regression on the same feature matrix.	0.5684	1.21×	0.02342	0.11561	0.01853
Random forest Forest regression; a learned similarity metric.	0.5740	1.35×	0.02334	0.11344	0.01619
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.5746	1.08×	0.02323	0.11256	0.01296
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.5744	0.94×	0.02324	0.11250	0.01259
Linear regression Ordinary least squares on the same features.	0.5766	0.94×	0.02333	0.11448	0.01600
Logistic regression Binomial logistic on the per-PA home run outcome.	0.5846	1.08×	0.02332	0.11370	0.01486
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.5721	1.21×	0.02329	0.11362	0.01393
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5843	1.21×	0.02337	0.11380	0.01667

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Shohei Ohtani 17.6%, Junior Caminero 17.1%, Jackson Chourio 15.2%, Jo Adell 14.7%, Teoscar Hernández 14.1%, Freddie Freeman 13.9%
Flat league	5/6	Junior Caminero 16.3%, Shohei Ohtani 16.0%, Jackson Chourio 15.2%, Jo Adell 14.0%, Freddie Freeman 13.5%, Will Smith 13.5%
KNN comps	5/6	Shohei Ohtani 17.6%, Junior Caminero 16.9%, Jackson Chourio 15.6%, Jo Adell 14.5%, Teoscar Hernández 14.2%, Jake Bauers 14.1%
SVR	5/6	Junior Caminero 18.6%, Shohei Ohtani 17.4%, Jackson Chourio 15.8%, Jo Adell 15.0%, Teoscar Hernández 14.1%, Jake Bauers 14.1%
Random forest	5/6	Shohei Ohtani 17.8%, Junior Caminero 17.5%, Jackson Chourio 15.9%, Jo Adell 14.6%, Teoscar Hernández 14.3%, Jake Bauers 14.1%
Core (KNN+SVR)	5/6	Shohei Ohtani 17.6%, Junior Caminero 16.9%, Jackson Chourio 15.6%, Jo Adell 14.5%, Teoscar Hernández 14.2%, Jake Bauers 14.1%
Forest blend	5/6	Shohei Ohtani 17.6%, Junior Caminero 17.0%, Jackson Chourio 15.6%, Jo Adell 14.5%, Teoscar Hernández 14.2%, Jake Bauers 14.1%
Linear regression	5/6	Shohei Ohtani 17.5%, Junior Caminero 17.1%, Jackson Chourio 15.6%, Jo Adell 14.6%, Will Smith 14.3%, Teoscar Hernández 14.3%
Logistic regression	5/6	Shohei Ohtani 17.9%, Junior Caminero 17.5%, Jackson Chourio 15.7%, Jo Adell 15.0%, Teoscar Hernández 14.3%, Will Smith 14.1%
Neural net (torch)	5/6	Shohei Ohtani 17.9%, Junior Caminero 17.6%, Jackson Chourio 15.8%, Jo Adell 15.0%, Teoscar Hernández 14.3%, Will Smith 14.2%
XGBoost	5/6	Shohei Ohtani 18.1%, Junior Caminero 18.0%, Jackson Chourio 16.0%, Jo Adell 14.8%, Jake Bauers 14.2%, Teoscar Hernández 14.2%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	Shohei Ohtani LAD slate	**17.6%**	16.0%	17.6%	17.4%	17.8%	17.6%	17.6%	17.5%	17.9%	
2	Junior Caminero TB slate	**17.1%**	16.3%	16.9%	18.6%	17.5%	16.9%	17.0%	17.1%	17.5%	
3	Jackson Chourio MIL slate	**15.2%**	15.2%	15.6%	15.8%	15.9%	15.6%	15.6%	15.6%	15.7%	
4	Jo Adell CLE slate	**14.7%**	14.0%	14.5%	15.0%	14.6%	14.5%	14.5%	14.6%	15.0%	
5	Teoscar Hernández LAD	**14.1%**	13.3%	14.2%	14.1%	14.3%	14.2%	14.2%	14.3%	14.3%	
6	Freddie Freeman LAD	**13.9%**	13.5%	13.7%	13.5%	13.6%	13.7%	13.7%	13.9%	13.6%	
7	Jake Bauers MIL	**13.9%**	13.0%	14.1%	14.1%	14.1%	14.1%	14.1%	13.9%	13.9%	
8	Will Smith LAD	**13.7%**	13.5%	13.8%	13.6%	13.8%	13.8%	13.8%	14.3%	14.1%	
9	Jonathan Aranda TB	**13.7%**	12.8%	13.8%	13.3%	13.4%	13.8%	13.8%	13.7%	13.4%	
10	Garrett Mitchell MIL	**13.4%**	12.8%	13.2%	12.1%	12.5%	13.2%	13.1%	12.6%	12.4%	
11	William Contreras MIL	**13.0%**	12.9%	13.0%	12.9%	13.0%	13.0%	13.0%	12.9%	12.9%	
12	Yandy Díaz TB	**13.0%**	13.4%	13.0%	13.4%	13.4%	13.0%	13.0%	13.3%	13.0%	
13	Brice Turang MIL	**12.8%**	13.1%	12.5%	12.8%	12.5%	12.5%	12.5%	12.2%	12.1%	
14	Mookie Betts LAD	**12.8%**	12.8%	12.7%	13.1%	12.6%	12.7%	12.7%	12.9%	12.9%	
15	Victor Mesa Jr. TB	**11.7%**	11.3%	11.8%	12.3%	11.9%	11.8%	11.8%	11.9%	11.9%	
16	Christian Yelich MIL	**11.7%**	11.9%	12.2%	11.3%	11.9%	12.2%	12.2%	12.2%	11.7%	
17	Andy Pages LAD	**10.5%**	10.1%	10.3%	10.1%	10.1%	10.3%	10.3%	10.0%	9.9%	
18	Nathaniel Lowe CLE	**10.4%**	10.1%	10.4%	10.2%	10.4%	10.4%	10.4%	10.3%	10.2%	
19	Chase DeLauter CLE	**10.3%**	10.4%	10.2%	10.1%	10.2%	10.2%	10.2%	10.0%	9.9%	
20	Angel Martínez CLE	**10.2%**	10.3%	10.1%	10.4%	10.1%	10.1%	10.1%	10.0%	10.1%	
21	Enrique Hernández LAD	**10.2%**	10.1%	10.3%	9.9%	10.2%	10.3%	10.3%	10.1%	9.7%	
22	Kyle Tucker LAD	**10.2%**	10.3%	10.3%	10.1%	10.2%	10.3%	10.3%	10.1%	9.9%	
23	Patrick Bailey CLE	**10.1%**	10.0%	10.5%	10.3%	10.1%	10.5%	10.5%	10.3%	9.9%	
24	José Ramírez CLE	**9.7%**	9.6%	9.5%	9.6%	9.5%	9.5%	9.5%	9.8%	10.0%	
25	Travis Bazzana CLE	**9.5%**	9.6%	9.4%	9.2%	9.6%	9.4%	9.4%	9.4%	9.1%	
26	Jonny DeLuca TB	**8.4%**	9.1%	8.2%	8.4%	7.6%	8.2%	8.2%	8.1%	8.5%	
27	Tommy Edman LAD	**7.1%**	7.8%	7.0%	7.1%	6.7%	7.0%	7.0%	7.2%	7.0%	
28	Liam Hicks TB	**7.1%**	8.2%	7.0%	8.0%	7.1%	7.0%	7.0%	7.5%	7.3%	

	HITTER		SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
29	**Brayan Rocchio**	CLE	**6.3%**	6.8%	6.1%	6.4%	6.0%	6.1%	6.1%	6.1%	6.1%	
30	**Joey Ortiz**	MIL	**6.1%**	7.1%	6.3%	6.3%	6.4%	6.3%	6.3%	6.2%	6.2%	
31	**Sal Frelick**	MIL	**6.0%**	7.6%	6.1%	6.0%	5.7%	6.1%	6.1%	6.0%	6.0%	
32	**Richie Palacios**	TB	**5.8%**	6.8%	5.8%	5.4%	5.1%	5.8%	5.7%	5.6%	5.6%	
33	**Cooper Pratt**	MIL	**5.6%**	7.3%	5.6%	4.9%	5.4%	5.6%	5.6%	5.0%	5.1%	
34	**Taylor Walls**	TB	**5.6%**	7.7%	5.8%	5.0%	4.9%	5.8%	5.7%	5.9%	5.9%	
35	**Steven Kwan**	CLE	**4.6%**	7.6%	5.3%	3.2%	4.0%	5.3%	5.2%	3.2%	4.6%	
36	**Chandler Simpson**	TB	**2.5%**	5.5%	3.4%	1.4%	1.6%	3.4%	3.3%	1.4%	2.9%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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